STUDENT ACTIVITY

Evolution and Adaptation — Activity

Activity: Machine Learning Approach and Data Resolution

Q: Read the business scenario below and describe how three different Machine Learning approaches would attempt to process the data to solve the specific departmental problems:

- One approach using a Supervised Regression model.
- One approach using a Supervised Classification model.
- One approach using an Unsupervised Clustering model.

Next, based on the provided text, describe the fundamental difference in data requirements and goals between Supervised and Unsupervised Learning in three sentences. Ensure you explicitly mention labeled versus unlabeled data.

The Scenario: A global e-commerce company wants to leverage its vast database of customer transactions and behavior to improve its business intelligence. First, the sales department wants to analyze historical sales data to predict the exact continuous numerical value of future revenue for the upcoming quarter. Second, the cybersecurity team needs a system to predict whether a new, incoming transaction should be labeled as "Fraud" or "Legitimate" based on past examples of fraudulent activities. Finally, the marketing team wants to analyze a massive database of users without any predefined categories to group customers based on similar purchasing behaviors and discover hidden spending patterns.

Machine Learning Approach and Data Resolution

Supervised Regression Model

Approach	How it Processes the Data	Business Problem Solved
Supervised Regression	The model learns from **labeled historical sales data**, where the correct revenue values are already known. It identifies relationships between input features (such as previous sales, customer demand, and seasonal trends) and the target revenue value to predict future revenue.	Predict the **exact continuous numerical value** of the company's revenue for the upcoming quarter.

Supervised Classification Model

Approach	How it Processes the Data	Business Problem Solved
Supervised Classification	The model is trained using **labeled transaction data**, where each transaction is marked as **"Fraud"** or **"Legitimate."** It learns patterns from previous examples and classifies new incoming transactions into the correct category.	Detect whether a new transaction is **Fraud** or **Legitimate**, helping the cybersecurity team prevent financial fraud.

Unsupervised Clustering Model

Approach	How it Processes the Data	Business Problem Solved
Unsupervised Clustering	The model analyzes **unlabeled customer data** and automatically groups customers with similar purchasing behaviors without predefined categories. It discovers hidden patterns and similarities among users.	Segment customers into groups for **targeted marketing**, personalized recommendations, and customer behavior analysis.

Difference Between Supervised and Unsupervised Learning

Supervised Learning uses **labeled data**, where the correct output is already known, to train a model that can make predictions or classifications on new data. **Unsupervised Learning** uses **unlabeled data**, where no predefined output labels are available, and its goal is to discover hidden patterns, relationships, or groups within the data. Therefore, supervised learning focuses on **prediction and classification**, while unsupervised learning focuses on **pattern discovery and data grouping**.

Machine Learning Approaches

Machine Learning Approach	How It Processes the Data	Departmental Problem Solved
Supervised Regression	Learns from **labeled historical sales data** to predict future revenue values.	Predict the **continuous numerical value** of next quarter's revenue.
Supervised Classification	Learns from **labeled fraud and legitimate transaction data** to classify new transactions.	Identify whether a transaction is **Fraud** or **Legitimate**.
Unsupervised Clustering	Groups **unlabeled customer data** based on similar purchasing behavior.	Discover customer segments and hidden spending patterns for marketing.

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Supervised vs. Unsupervised Learning

- **Supervised Learning** uses **labeled data** to learn the relationship between inputs and known outputs for prediction or classification.
- **Unsupervised Learning** uses **unlabeled data** to identify hidden patterns, relationships, or clusters without predefined outputs.
- Supervised learning aims for **accurate prediction and classification**, while unsupervised learning aims for **data exploration, pattern discovery, and customer grouping**.